Tonmya_Prescriptions

2026-08-17

Some simple models based on Tonmya sales

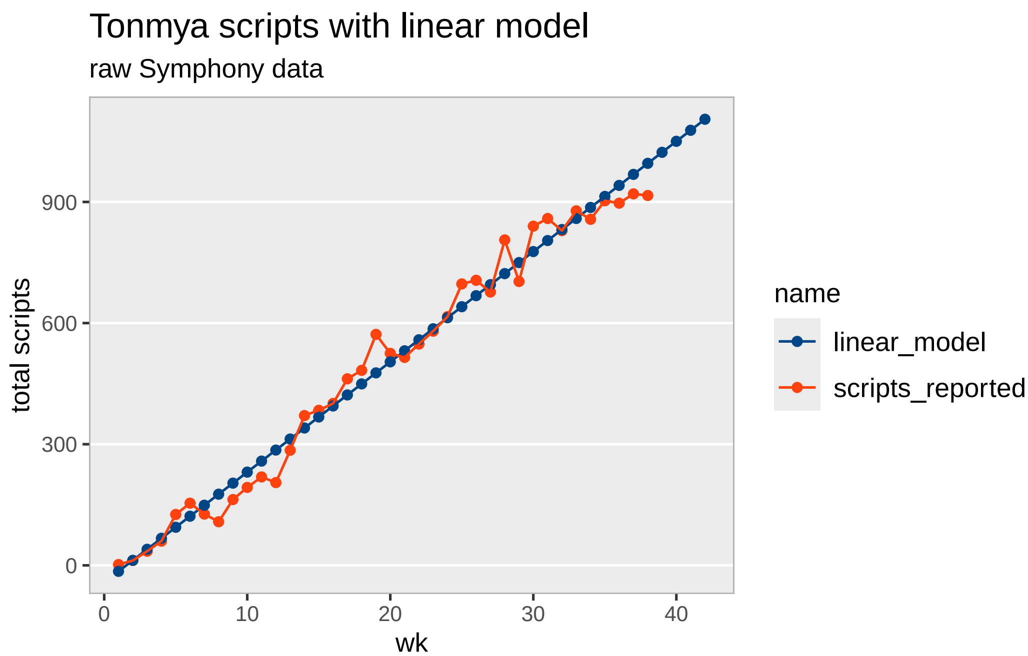
What are we doing here? - Just trying to get a grasp on the rate of increase and projecting out if that rate continues. It is not necessarily realistic to take it out as far as projections provide. Nothing here should be regarded as investment advice. I am not a financial analyst or advisor.

scripts Projection - at weekly rate - *linear model*

```
Call:
lm(formula = scripts ~ wk, data = df_scripts)

Residuals:
    Min      1Q  Median      3Q     Max
-80.539 -28.992  -3.747  31.437  95.287

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -42.1892    14.2108  -2.969  0.00529 **
wk           27.3106     0.6352  42.995  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 42.94 on 36 degrees of freedom
Multiple R-squared:  0.9809,    Adjusted R-squared:  0.9804
F-statistic:  1849 on 1 and 36 DF,  p-value: < 2.2e-16
```

Scripts, Linear Model

wk	weekly_seq	total reported	linear predicted	annual
1	2025-11-14	2	-14.9	1,420
2	2025-11-21	12	12.4	2,840
3	2025-11-28	35	39.7	4,260
4	2025-12-05	60	67.1	5,681
5	2025-12-12	126	94.4	7,101
6	2025-12-19	154	121.7	8,521
7	2025-12-26	127	149.0	9,941
8	2026-01-02	108	176.3	11,361
9	2026-01-09	163	203.6	12,781
10	2026-01-16	193	230.9	14,202
11	2026-01-23	219	258.2	15,622
12	2026-01-30	205	285.5	17,042
13	2026-02-06	285	312.8	18,462
14	2026-02-13	371	340.2	19,882
15	2026-02-20	384	367.5	21,302
16	2026-02-27	401	394.8	22,722
17	2026-03-06	462	422.1	24,143
18	2026-03-13	483	449.4	25,563
19	2026-03-20	572	476.7	26,983
20	2026-03-27	525	504.0	28,403
21	2026-04-03	515	531.3	29,823
22	2026-04-10	548	558.6	31,243
23	2026-04-17	580	586.0	32,664
24	2026-04-24	616	613.3	34,084
25	2026-05-01	697	640.6	35,504
26	2026-05-08	706	667.9	36,924
27	2026-05-15	677	695.2	38,344
28	2026-05-22	806	722.5	39,764
29	2026-05-29	703	749.8	41,184
30	2026-06-05	840	777.1	42,605
31	2026-06-12	859	804.4	44,025
32	2026-06-19	829	831.8	45,445
33	2026-06-26	878	859.1	46,865
34	2026-07-03	857	886.4	48,285
35	2026-07-10	903	913.7	49,705
36	2026-07-17	897	941.0	51,126
37	2026-07-24	920	968.3	52,546
38	2026-07-31	916	995.6	53,966
39	2026-08-07	NA	1,022.9	55,386
40	2026-08-14	NA	1,050.2	56,806
41	2026-08-21	NA	1,077.5	58,226
42	2026-08-28	NA	1,104.9	59,646



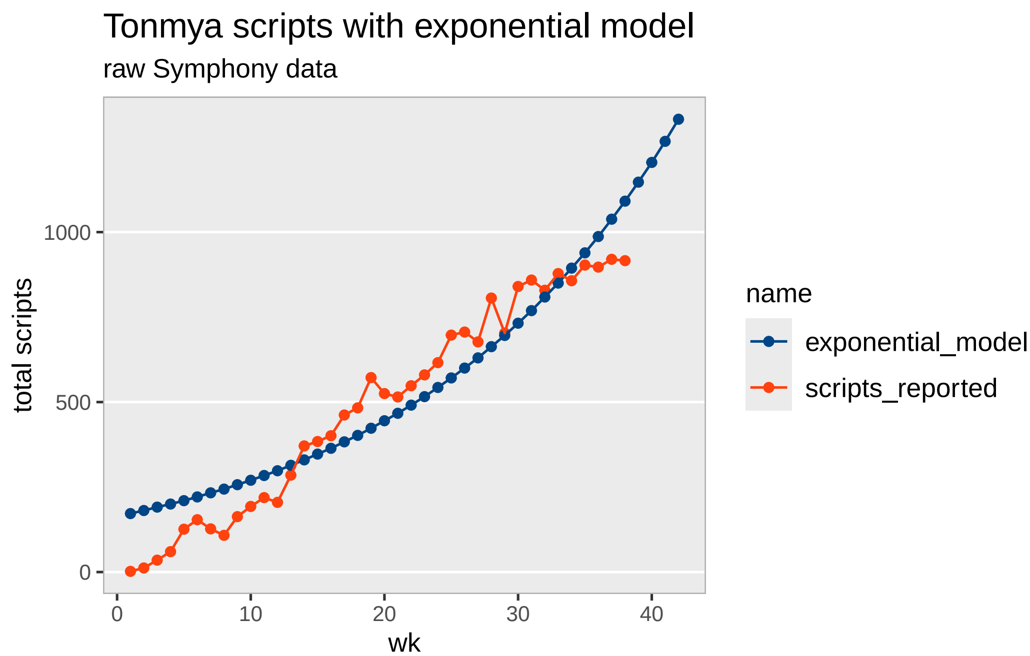

```
[1] 0.7349111
```

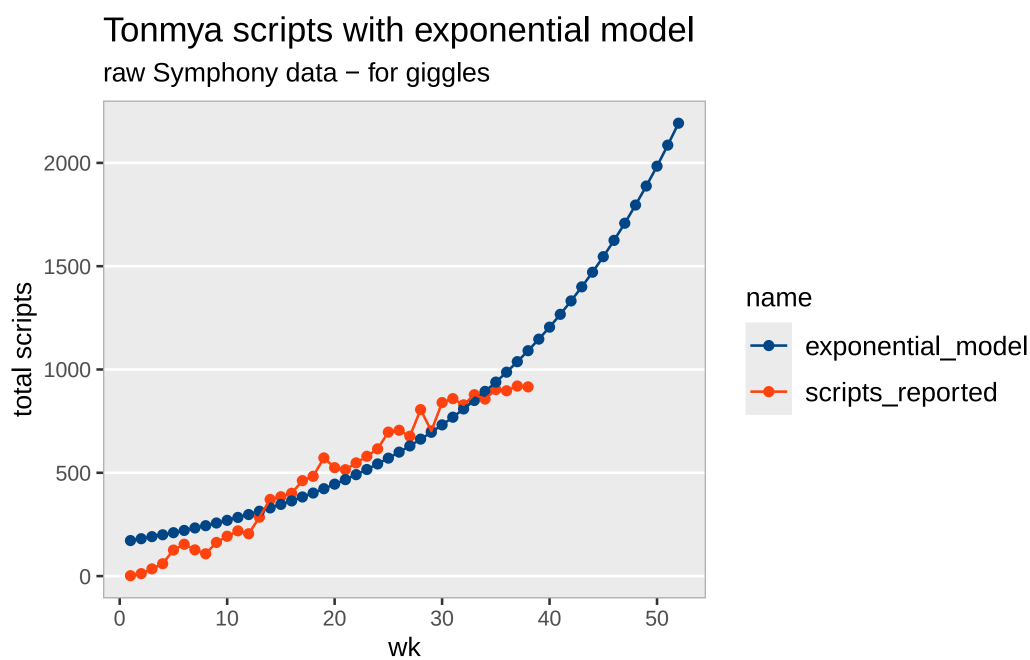
Scripts Projection - at weekly growth rate - *exponential model*

```
Formula: scripts ~ a * (1 + r)^wk

Parameters:
   Estimate Std. Error t value Pr(>|t|)
a 164.04030   17.48621   9.381 3.33e-11 ***
r   0.05112    0.00368  13.893 4.92e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 99.78 on 36 degrees of freedom

Number of iterations to convergence: 11
Achieved convergence tolerance: 5.669e-06
```

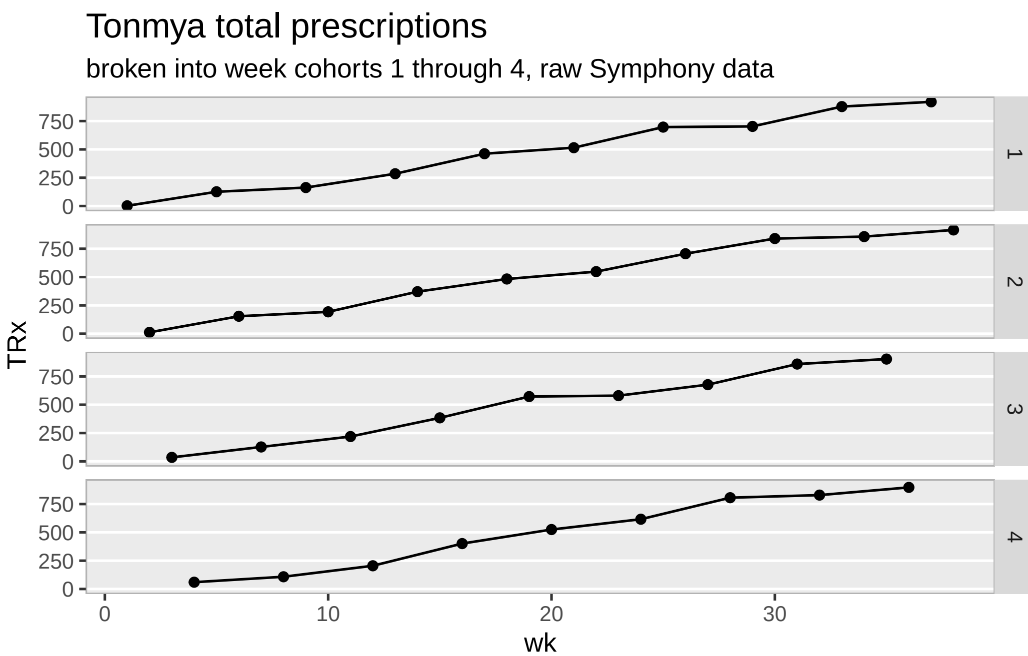
Scripts, Exponential Model

wk	weekly_seq	total reported	exp predicted	annual
1	2025-11-14	2	172	8,966
2	2025-11-21	12	181	9,425
3	2025-11-28	35	191	9,906
4	2025-12-05	60	200	10,413
5	2025-12-12	126	210	10,945
6	2025-12-19	154	221	11,505
7	2025-12-26	127	233	12,093
8	2026-01-02	108	244	12,711
9	2026-01-09	163	257	13,361
10	2026-01-16	193	270	14,044
11	2026-01-23	219	284	14,761
12	2026-01-30	205	298	15,516
13	2026-02-06	285	314	16,309
14	2026-02-13	371	330	17,143
15	2026-02-20	384	347	18,019
16	2026-02-27	401	364	18,941
17	2026-03-06	462	383	19,909
18	2026-03-13	483	402	20,927
19	2026-03-20	572	423	21,996
20	2026-03-27	525	445	23,121
21	2026-04-03	515	467	24,303
22	2026-04-10	548	491	25,545
23	2026-04-17	580	516	26,851
24	2026-04-24	616	543	28,224
25	2026-05-01	697	571	29,666
26	2026-05-08	706	600	31,183
27	2026-05-15	677	630	32,777
28	2026-05-22	806	663	34,452
29	2026-05-29	703	696	36,214
30	2026-06-05	840	732	38,065
31	2026-06-12	859	769	40,011
32	2026-06-19	829	809	42,056
33	2026-06-26	878	850	44,206
34	2026-07-03	857	894	46,466
35	2026-07-10	903	939	48,841
36	2026-07-17	897	987	51,338
37	2026-07-24	920	1038	53,962
38	2026-07-31	916	1091	56,721
39	2026-08-07	NA	1147	59,621
40	2026-08-14	NA	1205	62,668
41	2026-08-21	NA	1267	65,872
42	2026-08-28	NA	1332	69,239





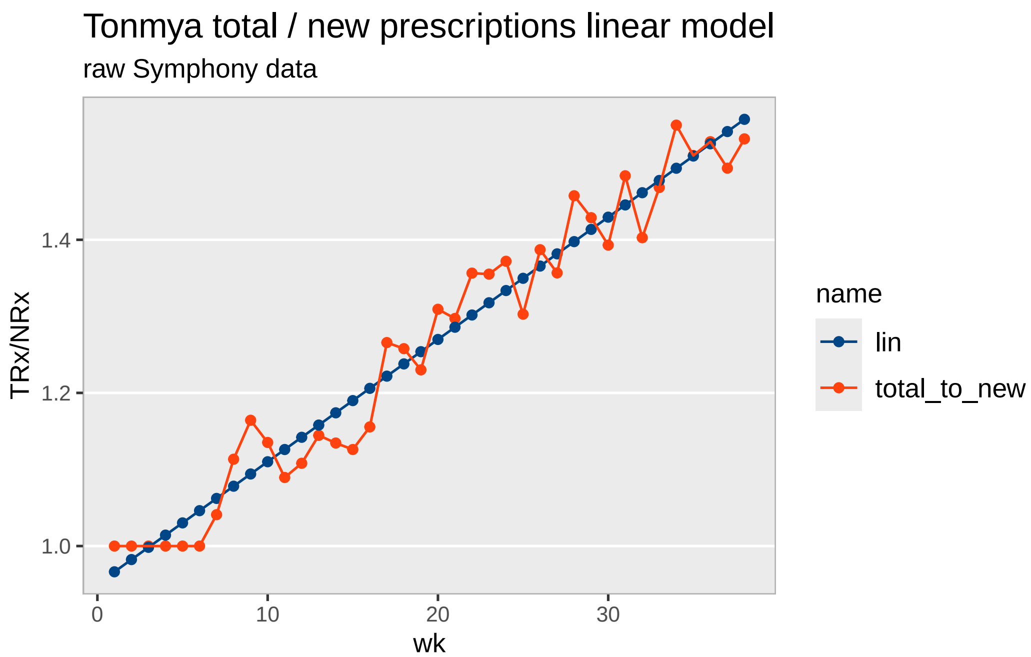
TRx Total Prescriptions by Weekly Cohorts



total / new scripts at weekly rate - ***linear model***

```
Call:
lm(formula = total_to_new ~ wk, data = trx_nrx_df)

Residuals:
      Min        1Q    Median        3Q       Max
-0.063867 -0.033025  0.001167  0.034829  0.070143

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) 0.9504051  0.0126669   75.03   <2e-16 ***
wk          0.0159708  0.0005662   28.21   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.03827 on 36 degrees of freedom
Multiple R-squared:  0.9567,    Adjusted R-squared:  0.9555
F-statistic: 795.6 on 1 and 36 DF,  p-value: < 2.2e-16
```

wk	scripts_reported
1	
1	2
5	126
9	163
13	285
17	462
21	515
25	697
29	703
33	878
37	920
2	
2	12
6	154
10	193
14	371
18	483
22	548
26	706
30	840
34	857
38	916
3	
3	35
7	127
11	219
15	384
19	572
23	580
27	677
31	859
35	903
4	
4	60
8	108
12	205
16	401
20	525
24	616
28	806
32	829
36	897

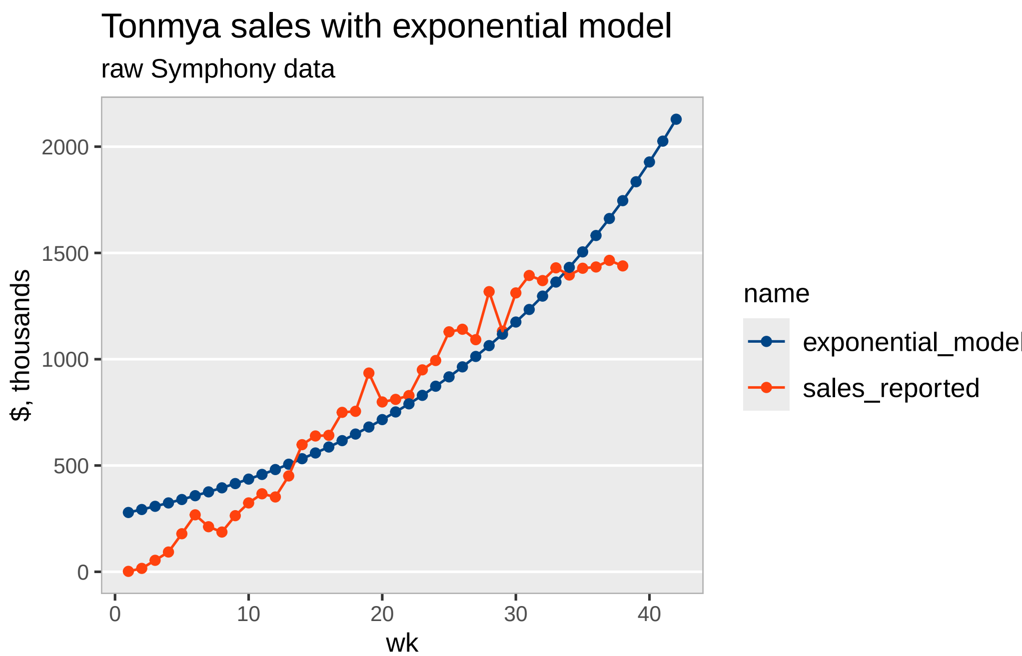


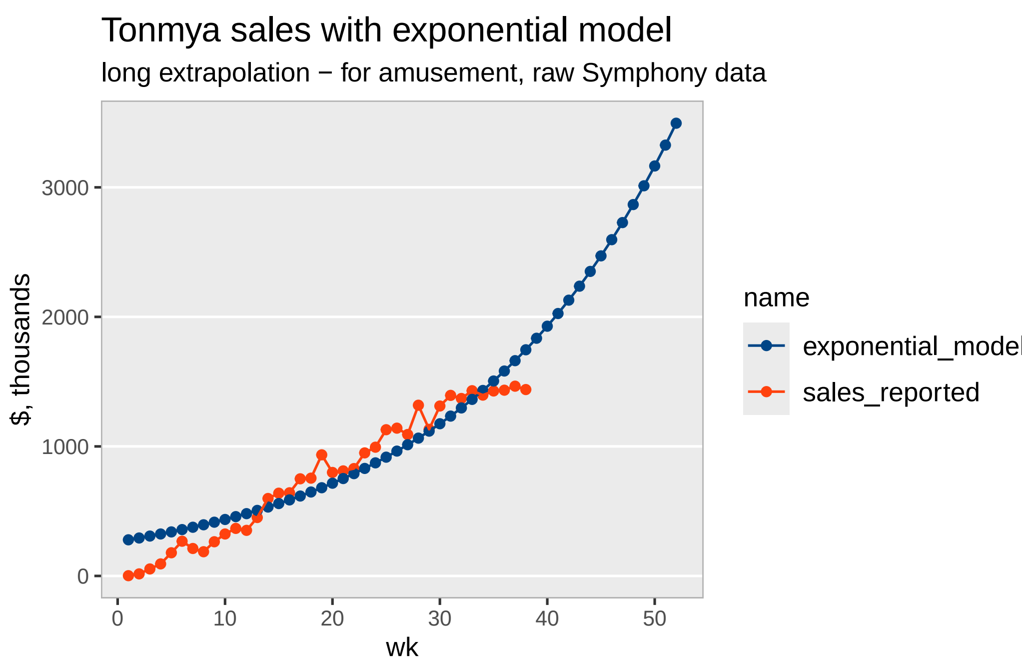
Sales Projection - at weekly growth rate - *exponential model*

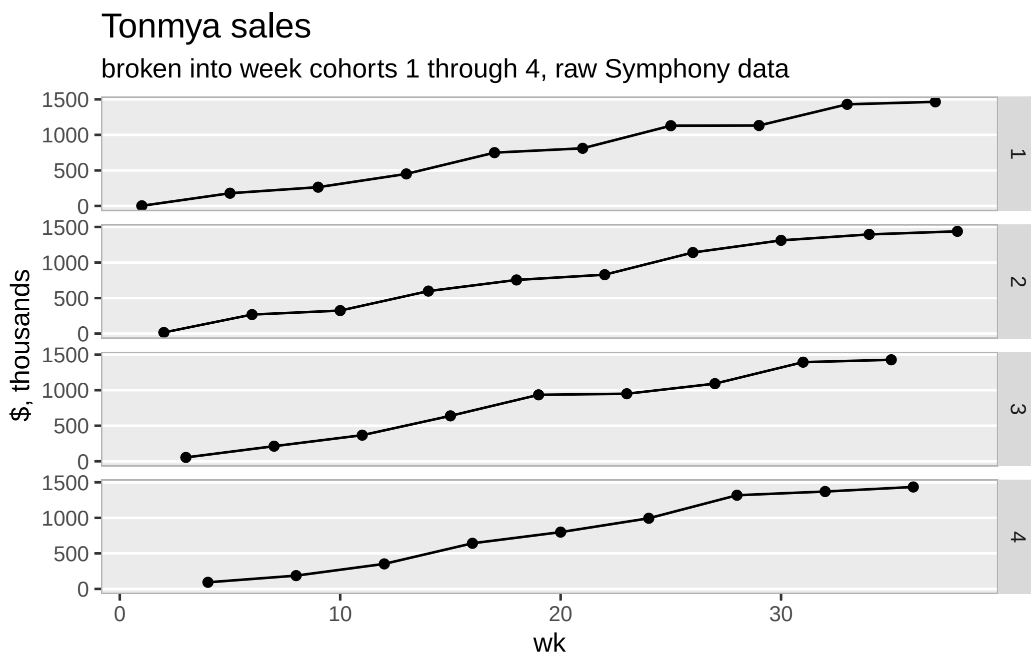
```
Formula: sales ~ a * (1 + r)^wk

Parameters:
   Estimate Std. Error t value Pr(>|t|)
a 2.657e+02  2.863e+01    9.28 4.40e-11 ***
r 5.080e-02  3.722e-03   13.65 8.45e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 162.6 on 36 degrees of freedom

Number of iterations to convergence: 13
Achieved convergence tolerance: 2.274e-06
```

Exponential Model

wk	weekly_seq	sales, thousands	exp, thousands	annual, M
1	2025-11-14	$2	$279	$14.5
2	2025-11-21	$16	$293	$15.3
3	2025-11-28	$54	$308	$16.0
4	2025-12-05	$93	$324	$16.8
5	2025-12-12	$179	$340	$17.7
6	2025-12-19	$268	$358	$18.6
7	2025-12-26	$212	$376	$19.5
8	2026-01-02	$187	$395	$20.5
9	2026-01-09	$264	$415	$21.6
10	2026-01-16	$324	$436	$22.7
11	2026-01-23	$367	$458	$23.8
12	2026-01-30	$352	$481	$25.0
13	2026-02-06	$451	$506	$26.3
14	2026-02-13	$598	$532	$27.6
15	2026-02-20	$639	$559	$29.1
16	2026-02-27	$642	$587	$30.5
17	2026-03-06	$750	$617	$32.1
18	2026-03-13	$755	$648	$33.7
19	2026-03-20	$935	$681	$35.4
20	2026-03-27	$799	$716	$37.2
21	2026-04-03	$811	$752	$39.1
22	2026-04-10	$829	$790	$41.1
23	2026-04-17	$950	$830	$43.2
24	2026-04-24	$994	$873	$45.4
25	2026-05-01	$1,129	$917	$47.7
26	2026-05-08	$1,141	$964	$50.1
27	2026-05-15	$1,092	$1,013	$52.7
28	2026-05-22	$1,318	$1,064	$55.3
29	2026-05-29	$1,132	$1,118	$58.1
30	2026-06-05	$1,312	$1,175	$61.1
31	2026-06-12	$1,394	$1,234	$64.2
32	2026-06-19	$1,370	$1,297	$67.5
33	2026-06-26	$1,430	$1,363	$70.9
34	2026-07-03	$1,396	$1,432	$74.5
35	2026-07-10	$1,428	$1,505	$78.3
36	2026-07-17	$1,434	$1,582	$82.2
37	2026-07-24	$1,465	$1,662	$86.4
38	2026-07-31	$1,439	$1,746	$90.8
39	2026-08-07	NA	$1,835	$95.4
40	2026-08-14	NA	$1,928	$100.3
41	2026-08-21	NA	$2,026	$105.4
42	2026-08-28	NA	$2,129	$110.7


Tonmya sales with exponential model
raw Symphony data
$, thousands
2000
1500
1000
500
0
0
10
20
30
40
wk
name
exponential_model
sales_reported


Tonmya sales with exponential model
long extrapolation − for amusement, raw Symphony data
$, thousands
3000
2000
1000
0
0
10
20
30
40
50
wk
name
exponential_model
sales_reported

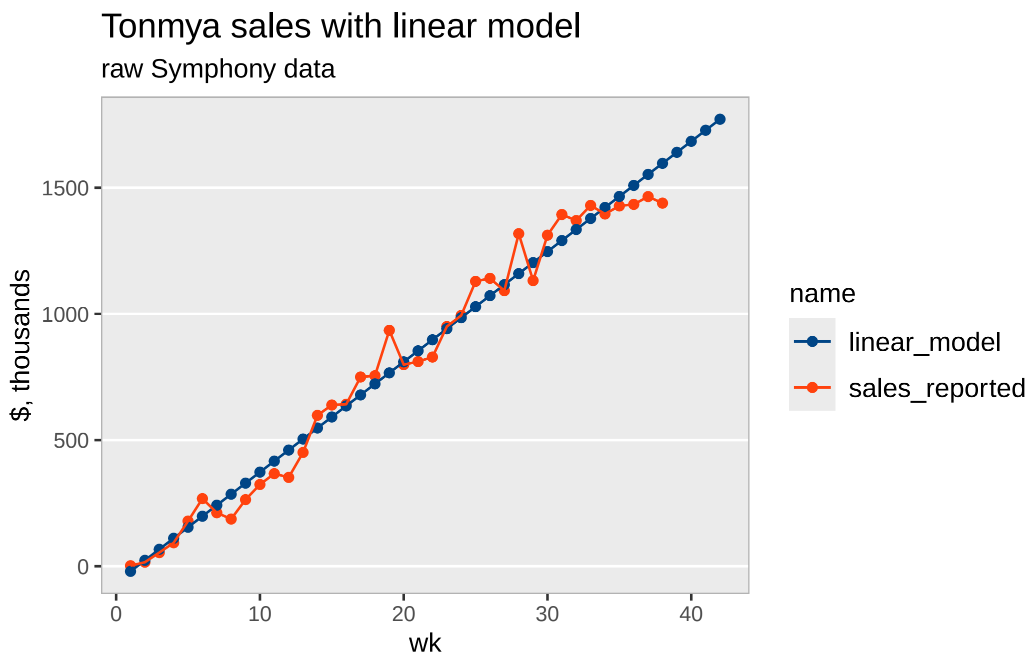


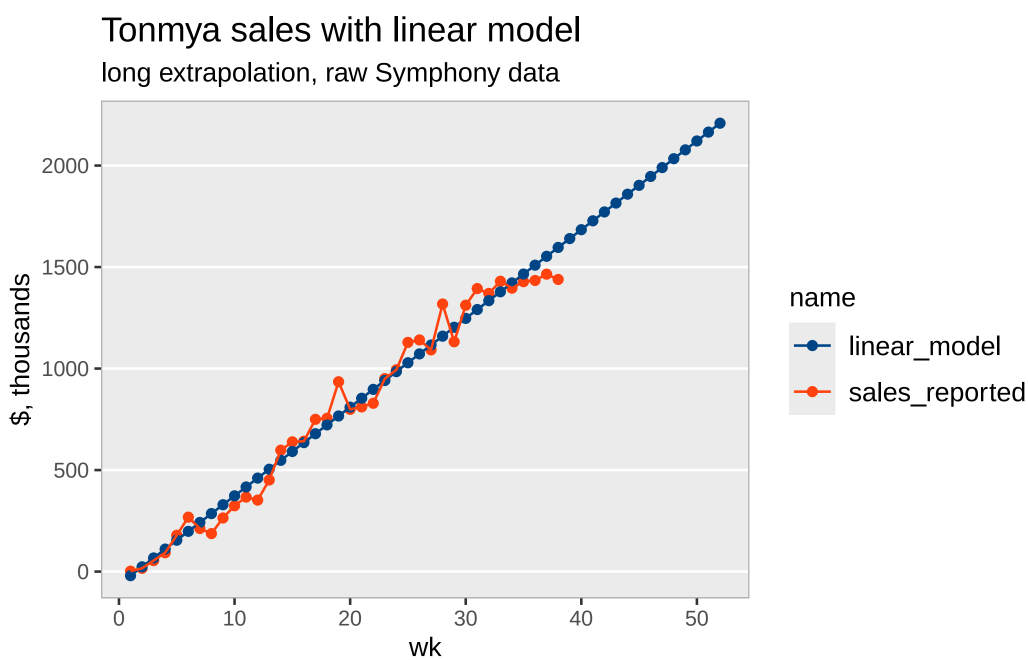
Sales Projection - at weekly rate - *linear model*

```
Call:
lm(formula = sales ~ wk, data = df_sales)

Residuals:
     Min       1Q   Median       3Q      Max 
-157.692  -49.532   -9.208   49.507  168.667 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)   -64.03      24.15  -2.651   0.0119 *  
wk             43.70       1.08  40.481   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 72.98 on 36 degrees of freedom
Multiple R-squared:  0.9785,    Adjusted R-squared:  0.9779 
F-statistic:  1639 on 1 and 36 DF,  p-value: < 2.2e-16
```

Linear Model

wk	weekly_seq	sales, thousands	linear, thousands	annual, M
1	2025-11-14	$2	-$20	$2.3
2	2025-11-21	$16	$23	$4.5
3	2025-11-28	$54	$67	$6.8
4	2025-12-05	$93	$111	$9.1
5	2025-12-12	$179	$154	$11.4
6	2025-12-19	$268	$198	$13.6
7	2025-12-26	$212	$242	$15.9
8	2026-01-02	$187	$286	$18.2
9	2026-01-09	$264	$329	$20.5
10	2026-01-16	$324	$373	$22.7
11	2026-01-23	$367	$417	$25.0
12	2026-01-30	$352	$460	$27.3
13	2026-02-06	$451	$504	$29.5
14	2026-02-13	$598	$548	$31.8
15	2026-02-20	$639	$592	$34.1
16	2026-02-27	$642	$635	$36.4
17	2026-03-06	$750	$679	$38.6
18	2026-03-13	$755	$723	$40.9
19	2026-03-20	$935	$766	$43.2
20	2026-03-27	$799	$810	$45.5
21	2026-04-03	$811	$854	$47.7
22	2026-04-10	$829	$897	$50.0
23	2026-04-17	$950	$941	$52.3
24	2026-04-24	$994	$985	$54.5
25	2026-05-01	$1,129	$1,029	$56.8
26	2026-05-08	$1,141	$1,072	$59.1
27	2026-05-15	$1,092	$1,116	$61.4
28	2026-05-22	$1,318	$1,160	$63.6
29	2026-05-29	$1,132	$1,203	$65.9
30	2026-06-05	$1,312	$1,247	$68.2
31	2026-06-12	$1,394	$1,291	$70.4
32	2026-06-19	$1,370	$1,334	$72.7
33	2026-06-26	$1,430	$1,378	$75.0
34	2026-07-03	$1,396	$1,422	$77.3
35	2026-07-10	$1,428	$1,466	$79.5
36	2026-07-17	$1,434	$1,509	$81.8
37	2026-07-24	$1,465	$1,553	$84.1
38	2026-07-31	$1,439	$1,597	$86.4
39	2026-08-07	NA	$1,640	$88.6
40	2026-08-14	NA	$1,684	$90.9
41	2026-08-21	NA	$1,728	$93.2
42	2026-08-28	NA	$1,772	$95.4





Refills - at weekly rate - *linear model*

```
Call:
lm(formula = refills ~ wk, data = trx_nrx_df)
```

```
Residuals:
    Min      1Q  Median      3Q     Max 
-40.429 -15.528  -4.322  17.748  55.073 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  -64.966      8.232  -7.892  2.3e-09 ***
wk             9.893      0.368  26.887  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 24.87 on 36 degrees of freedom
Multiple R-squared:  0.9526,    Adjusted R-squared:  0.9512 
F-statistic: 722.9 on 1 and 36 DF,  p-value: < 2.2e-16
```